

OSLAB PRACTICAL-10 PROGRAMS & OUTPUT

Date: 03-10-26

Part – A Output:

```
sainath@Sainath:~/OSLAB$ mkdir practical_10
sainath@Sainath:~/OSLAB$ cd practical_10
sainath@Sainath:~/OSLAB/practical_10$ echo "Hello from inode practical" > original.txt
sainath@Sainath:~/OSLAB/practical_10$ stat original.txt
  File: original.txt
  Size: 27          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 12055         Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/ sainath)   Gid: ( 1000/ sainath)
Access: 2026-10-06 05:11:13.238816861 +0000
Modify: 2026-10-06 05:11:13.238816861 +0000
Change: 2026-10-06 05:11:13.238816861 +0000
 Birth: 2026-10-06 05:11:13.238816861 +0000
sainath@Sainath:~/OSLAB/practical_10$ ln original.txt hardlink.txt
sainath@Sainath:~/OSLAB/practical_10$ ls -li original.txt hardlink.txt
12055 -rw-r--r--  2 sainath sainath 27 Oct  6 05:11 hardlink.txt
12055 -rw-r--r--  2 sainath sainath 27 Oct  6 05:11 original.txt
sainath@Sainath:~/OSLAB/practical_10$ ln -s original.txt symlink.txt
sainath@Sainath:~/OSLAB/practical_10$ ls -li original.txt hardlink.txt symlink.txt
12055 -rw-r--r--  2 sainath sainath 27 Oct  6 05:11 hardlink.txt
12055 -rw-r--r--  2 sainath sainath 27 Oct  6 05:11 original.txt
1515 lrwxrwxrwx  1 sainath sainath 12 Oct  6 05:12 symlink.txt -> original.txt
sainath@Sainath:~/OSLAB/practical_10$ find . -inum 123456
sainath@Sainath:~/OSLAB/practical_10$ rm original.txt
sainath@Sainath:~/OSLAB/practical_10$ cat hardlink.txt
Hello from inode practical
sainath@Sainath:~/OSLAB/practical_10$ cat symlink.txt
cat: symlink.txt: No such file or directory
sainath@Sainath:~/OSLAB/practical_10$ gedit program1.c
sainath@Sainath:~/OSLAB/practical_10$ gcc program1.c -o program1
sainath@Sainath:~/OSLAB/practical_10$ ./program1
Data written using mmap: Hello from mmap!
```

Program-1 CODE & OUTPUT:

```
sainath@Sainath:~/OSLAB/practical_10$ cat program1.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <string.h>

int main() {
    int fd;
    char *data;
    char message[] = "Hello from mmap!";

    fd = open("mmap.txt", O_RDWR | O_CREAT, 0644);

    if (fd < 0) {
        perror("open");
        return 1;
    }

    ftruncate(fd, 100);

    data = mmap(NULL, 100,
                PROT_READ | PROT_WRITE,
                MAP_SHARED, fd, 0);

    if (data == MAP_FAILED) {
        perror("mmap");
        return 1;
    }

    strcpy(data, message);

    printf("Data written using mmap: %s\n", data);

    munmap(data, 100);
    close(fd);

    return 0;
}
sainath@Sainath:~/OSLAB/practical_10$ ./program1
Data written using mmap: Hello from mmap!
```

History:

```
575 mkdir practical_10
576 cd practical_10
577 echo "Hello from inode practical" > original.txt
578 stat original.txt
579 ln original.txt hardlink.txt
580 ls -li original.txt hardlink.txt
581 ln -s original.txt symlink.txt
582 ls -li original.txt hardlink.txt symlink.txt
583 find . -inum 123456
584 rm original.txt
585 cat hardlink.txt
586 cat symlink.txt
587 gedit program1.c
588 gcc program1.c -o program1
589 ./program1
590 cat program1.c
591 ./program1.c
592 cat program1.c
593 ./program1
594 history
```